

Def. Let F be a field. Given a polynomial $f(x) \in F[x]$ over F , the Extension $E \stackrel{\text{def}}{=} F(\alpha_1, \dots, \alpha_n)$ where $\alpha_1, \dots, \alpha_n \in \overline{F}$ are roots of f .

is called the splitting field of $f(x)$.

Further, the collection of polynomials $\{f_i(x) \mid i \in I\}$, the intersection of splitting fields of f_i is also called the splitting field.

Thm. Let $\varphi: F \rightarrow F'$ be an isomorphism of fields.

Given $f(x) = a_n x^n + \dots + a_1 x + a_0 \in F[x]$, let $f'(x) = \varphi(a_n) x^n + \dots + \varphi(a_0) \in F'[x]$.

Let E be the splitting field for $f(x)$ over F ,

and E' be the splitting field for $f'(x)$ over F' .

Then, φ can be extended on E to E' , isomorphically.